

Newborough Warren Coastal Aquifer

Summary of Hydrogeological Analysis

A plain-text summary of Hollingham (2026), prepared March 2026

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1. Study at a glance

Newborough Warren is a 1,300-hectare Special Area of Conservation on the Isle of Anglesey, Wales, designated for its humid dune slack communities. These communities depend on a shallow water table that rises into the slack floor each winter and stays close to the surface through summer, and have been in documented decline for six decades. This study applies a reproducible analytical pipeline to 21 years of monthly dipwell records (April 2005–February 2026) from a 69-well reference network, integrating nine complementary analyses: hierarchical clustering, Pearson affinity analysis, state-space modelling, boundary intercept auditing, comparative model benchmarking, critical rainfall threshold forecasting, an expanded clear-felling BACI experiment, dune scraping analysis, cluster-level water balance decomposition, and water table fluctuation specific yield estimation.

The headline findings are that the dune field is split by its underlying geology into hydrologically distinct blocks behaving like a shallow pan (Eastern) and a deep sponge (Western), with a canopy sponge (Forest) overlaid on the northern margin; that summer water tables are deepening across every cluster — including clusters on drainage systems independent of the plantation — at a rate consistent with climate-driven warming rather than with progressive forest evapotranspiration as a site-wide driver; that the site runs a persistent monthly water balance deficit sustained by unmeasured flux from outside the dune field, whose source differs from cluster to cluster; and that the December 2017 clearfell — after correcting for climate variability and the upslope-draining effect of the prior 2015 scraping — produced a step lowering of the impact-zone water table that has continued to accumulate rather than recover. The net effect on summer minima, the ecologically decisive variable, was negative.

2. The hydrogeological framework

2.1 Six clusters, four that matter

Ward's hierarchical clustering on the correlation structure of monthly hydrographs ($k = 6$, silhouette coefficient = 0.41) identified six behavioural groups. Four carry the bulk of the analysis:

- **C1 — Eastern Block Lake-buffer** (8 wells). Shallow sand over glacial till and estuarine deposits, immediately adjacent to Llyn Rhos-Ddu. Flashy, low storage.
- **C2 — Eastern Block Mature Dune** (28 wells). The main body of the eastern aquifer. Similar response behaviour to C1 but with more storage.
- **C3 — Western Block Mature Dune** (22 wells). Deep clean dune sand over a broader platform. Buffered, slow, high storage capacity.

- **C4 — Forest** (9 wells). Open dune substrate (not a separate geology) beneath the Corsican pine plantation. Behaviour dominated by canopy interception rather than substrate.

C5 (Coastal, $n = 1$) and C6 (Lake, $n = 1$) are excluded from mechanistic interpretation because tidal exchange and near-fixed-head boundary conditions respectively violate the drainage-decay assumptions of the state-space model.

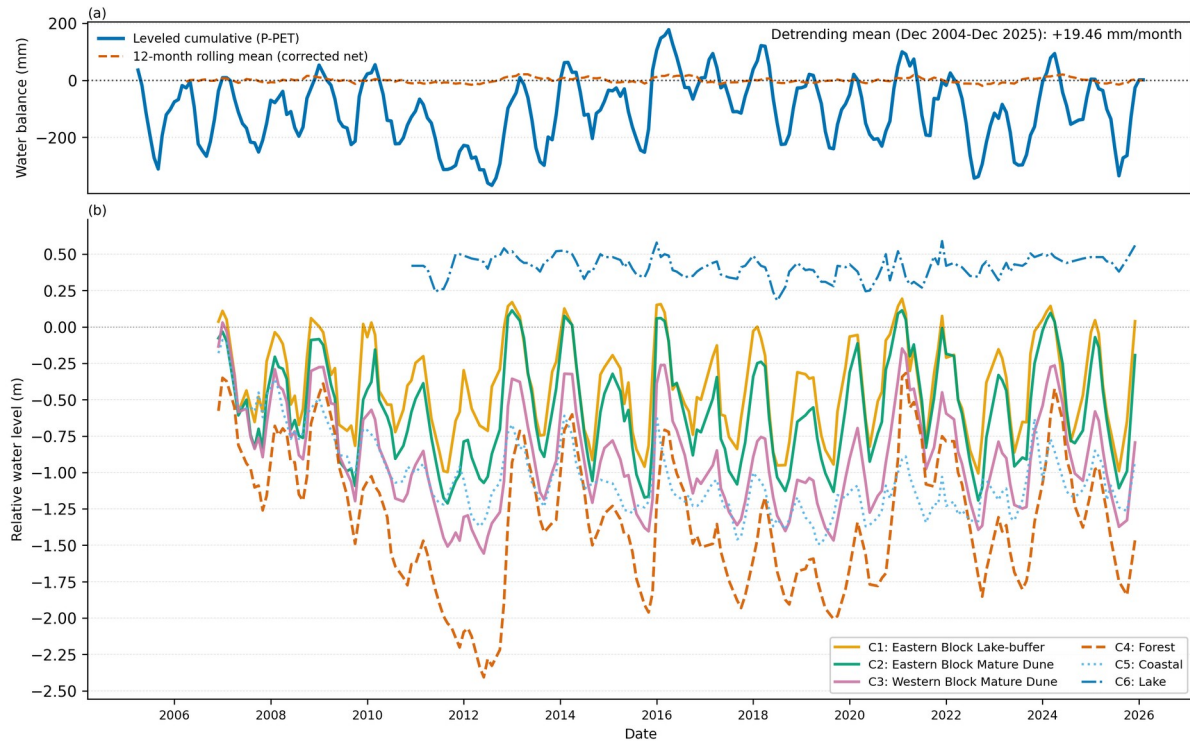


Figure 1. Cluster-mean relative water level hydrographs for the six hydrogeological clusters, April 2005–February 2026. Upper panel: levelled cumulative water balance ($P - PET$), detrended at the study-period mean of +19.46 mm/month; the 12-month rolling mean of the corrected net balance (orange dashed) confirms no long-run directional trend in climate forcing over the monitoring period. Lower panel: monthly cluster-mean water level relative to pipe-top datum, where zero approximates ground surface. C6 (Lake) is available from December 2010 only. Hollingham (2026), Figure 7.

2.2 Three coefficients that describe everything

For each cluster, a state-space model fits three coefficients to the monthly change in water table depth:

- **β_1 (recharge sensitivity)** — how much the water table rises per mm of rainfall.
- **β_2 (atmospheric draw)** — how much the water table falls per mm of potential evapotranspiration.
- **$-\beta_3$ (storage decay)** — the proportional rate at which stored water drains each month when no forcing is applied.

The Lumped Catchment Storage Coefficient $LCSC = 100/\beta_1$ converts recharge sensitivity into a management-friendly metric: the mm of rainfall required to raise the water table by 10 cm.

State-Space Mechanistic Signatures by Hydrogeological Cluster

Centroid fit: one SSM per cluster-mean hydrograph (matches 03_03_cluster_mechanistic_coefficients.csv)

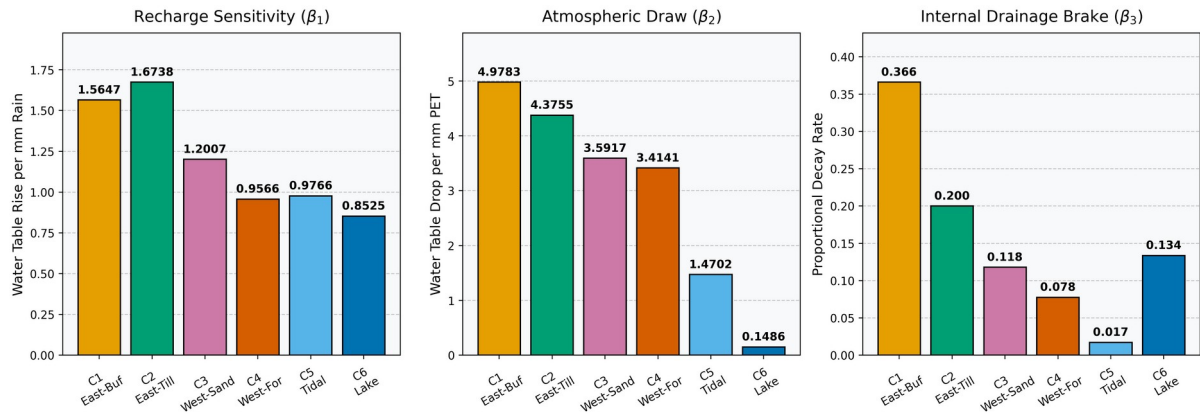


Figure 2. State-space mechanistic signatures by cluster. Eastern Block wells (C1, C2) have the highest recharge sensitivity and the fastest drainage; the Western Block (C3) responds more slowly but holds more water; the Forest (C4) sits at the low end of every coefficient, consistent with canopy interception smoothing both inputs and losses. Values from Hollingham (2026), Table 2 — centroid fit (matches Table 1 in this summary).

2.3 Cluster coefficients at a glance

Cluster	Label	β_1	β_2	$-\beta_3$	LCSC (%)
C1	Eastern Block Lake-buffer	1.56	4.98	0.366	63.9
C2	Eastern Block Mature Dune	1.67	4.38	0.200	59.7
C3	Western Block Mature Dune	1.20	3.59	0.118	83.3
C4	Forest	0.96	3.41	0.078	104.5

Table 1. Cluster mechanistic coefficients (Hollingham 2026, Table 2). LCSC > 100% at C4 reflects canopy interception reducing effective recharge, so more than 100 mm of gross rainfall is required to raise the water table by 10 cm — not an error.

3. State-space modelling outperforms the traditional approach

The state-space model (SSM) and a traditional linear regression (TLM) were fitted at every one of the 69 reference wells and benchmarked on iterative multi-step forecasting. The SSM's explicit parameterisation of storage decay transforms forecast skill: the median Nash–Sutcliffe Efficiency improvement over TLM is +0.586. Of the 69 wells, 68 return a positive NSE with the SSM, compared with only 33 with TLM. Put another way, 36 of the 69 traditional models perform worse than a climatological-mean forecast, versus only 1 of the 69 state-space models.

Spatial mapping of the SSM gain is informative in itself: the wells where the SSM gains most over TLM are precisely the boundary wells where unmeasured lateral flow matters, effectively delineating subsurface structure without invasive coring.

4. Every cluster runs a monthly water balance deficit

Cluster-level water balance decomposition using the SSM coefficients shows that every structural domain runs a persistent monthly deficit: losses exceed local recharge everywhere. The flux required to close this deficit — the water balance residual — is a directly computed quantity that averages 314–351 mm/year across clusters, compared with local rainfall recharge of only 99–173 mm/year. This is a structural feature of the aquifer. What is less well constrained, and requires more care to state, is the physical identity of the residual.

Positive intercepts from an independent boundary-intercept audit are concentrated at the northern forest margin and the lake-buffer sector — with CEH14 carrying the largest boundary subsidy in the network ($\alpha = +0.222$ m/month) — making ridge-derived lateral recharge the most parsimonious single candidate. However, two diagnostic time-structure tests run on the residuals (a cross-correlation lag analysis and a seasonal climatology analysis; Supplementary Notes S5 and S6) both returned null results, neither uniquely corroborating event-driven ridge transport nor unmodelled summer evaporation. The residual is therefore real and spatially structured, but its mechanistic decomposition cannot be pinned down from water-level data alone. The most defensible summary is that the residual likely reflects a combination of steady ridge baseflow, nonlinear soil-moisture storage dynamics the linear SSM does not capture, and minor β -coefficient miscalibration, in proportions that differ between clusters.

A single uniform ridge-to-site pathway is not supported by the network geometry. Three distinct pathways can be distinguished: (i) ridge → forest → sea, feeding C4 and forest-adjacent C3 wells — the pathway for which the evidence is strongest; (ii) ridge → Llyn Rhos-Ddu → C1 lake-buffer, in which the lake intercepts and integrates ridge recharge before releasing it gradually into the adjacent aquifer; and (iii) ridge → open warren → C2 and warren-interior C3, reaching the open dunes largely without passing through the forest but not instrumented close enough to the ridge to confirm the lateral inflow directly. These pathways are independent: an action affecting one does not automatically affect the others.

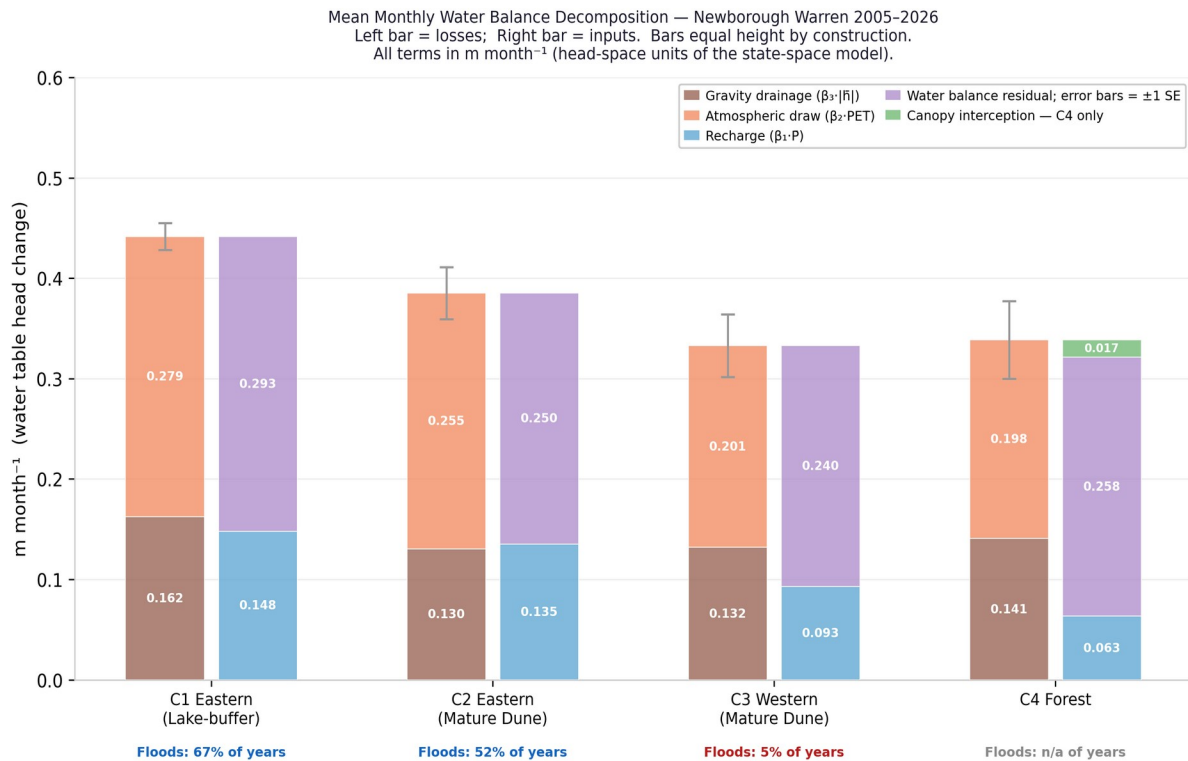


Figure 3. Volumetric annual water balance by cluster (Hollingham 2026, Table 3b / Figure 8b). Local rainfall recharge (blue) is consistently smaller than losses (orange + dark brown); the water balance residual (purple) is the flux required to close the annual deficit. Using conservative lower-bound S_y (0.08 for C1, 0.12 for C2–C4), the residual is 314–351 mm/year — about one third of mean annual rainfall. Under WTF upper-bound S_y (0.22–0.26), the residual would scale to 702–785 mm/year (79–88% of mean annual rainfall; Table 3d); this is considered implausibly large and is the basis for retaining the lower-bound S_y as the primary reference.

The implication for management is directional rather than prescriptive. Whatever its mechanistic composition, the residual is sourced from outside the dune field, and actions that alter the water balance upstream of each pathway — Llyn Rhos-Ddu itself, the ridge slopes, or cross-site drainage — could have consequences that propagate into the dipwell network only after substantial delay. Direct measurement at the source candidates is identified as a priority for future work (Section 11).

5. What the interventions actually did

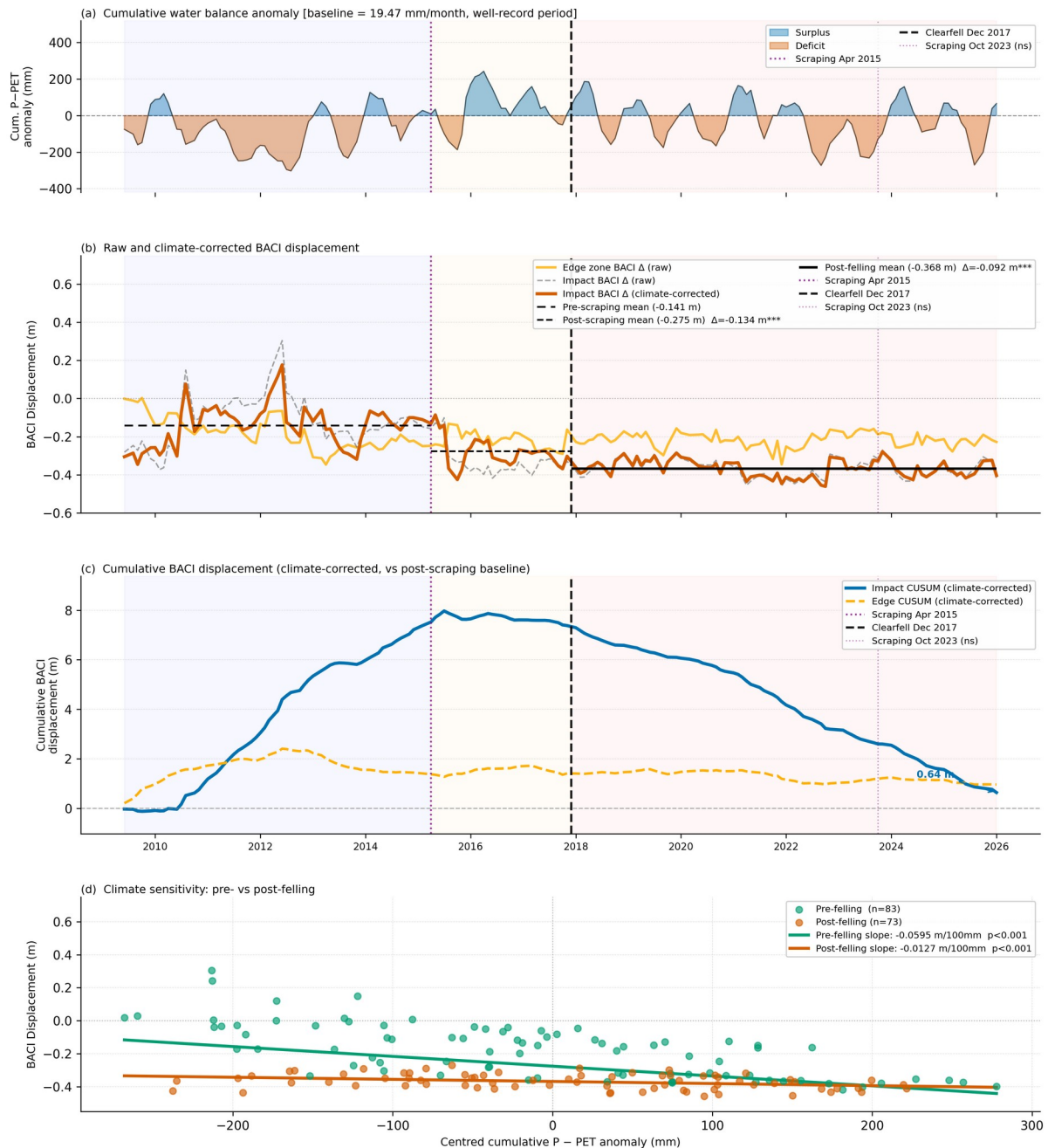
5.1 The December 2017 clearfell: signal is weaker than it looks

The ANCOVA-BACI model, built on an expanded 16-well experimental network and using a cumulative water balance anomaly as a climate covariate, explains 72.9% of monthly BACI displacement variance. Under this model the December 2017 clearfell imposed a climate-corrected step of -0.093 m on the core impact zone, on top of a -0.137 m step attributable to the prior April 2015 dune scraping at CEH36 (which sits downslope of the felling area and had already been drawing the upslope water table down) — giving a combined climate-corrected cost of -0.230 m relative to the pre-scraping baseline.

That headline step change is, however, an upper bound rather than a confirmed clearfell-specific signal. Three independent robustness checks (SSM residual normalisation, synthetic control, cluster transition testing) failed to recover a clearfell-specific effect once the scraping confound and the anomalously wet 2015–16 pre-felling baseline were accounted for. Post-felling increases in the storage decay coefficient ($-\beta_3$) appear at 13 of 14 paired wells including controls several hundred metres from the felling boundary, and are therefore attributed to SSM fitting artefacts of the transition between equilibria rather than genuine drainage acceleration.

What is genuine is the change in β_1 and β_2 . β_1 increased at 12 of 13 paired wells (canopy no longer intercepting rainfall) and β_2 increased at all 13 paired wells (canopy no longer shielding the water table from evaporative demand). Of these, the β_2 increase is the ecologically consequential one: it deepens summer minima more than any winter improvement raises winter maxima. By February 2026 the climate-corrected cumulative BACI displacement showed no sign of stabilising, and the climate sensitivity of the impact zone was reduced by 79.3% following canopy removal.

ANCOVA-BACI (Model 2): Climate- and scraping-corrected clearfell hydrological impact
 2015 scraping step = -0.134 m $p < 0.001$ | Clearfell step = -0.092 m $[-0.129, -0.056]$ $p < 0.001$ | Combined cost = -0.226 m $R^2 = 0.600$



† Oct 2023 scraping term: coef = -0.031 m , $p = 0.258$, $\Delta\text{AIC} = +0.66$ — not retained in final model.

Figure 4. ANCOVA-BACI analysis of the December 2017 clearfell (Hollingham 2026, Figure 22; $R^2 = 0.729$, $n = 155$). Panel (a) cumulative water balance anomaly ($P - PET - 19.47\text{ mm/month}$), showing the anomalously wet 2015–16 pre-clearfell period. Panel (b) raw (grey dashed) and climate-corrected (orange) BACI displacement for the core impact zone, with fitted era means — scraping step -0.137 m , clearfell step -0.093 m [95% CI $-0.131, -0.055$]. Panel (c) climate-corrected CUSUM; the cumulative post-felling deficit has not reversed the pre-intervention surplus by February 2026. Panel (d) climate-sensitivity scatter: 79.3% reduction following clearfell.

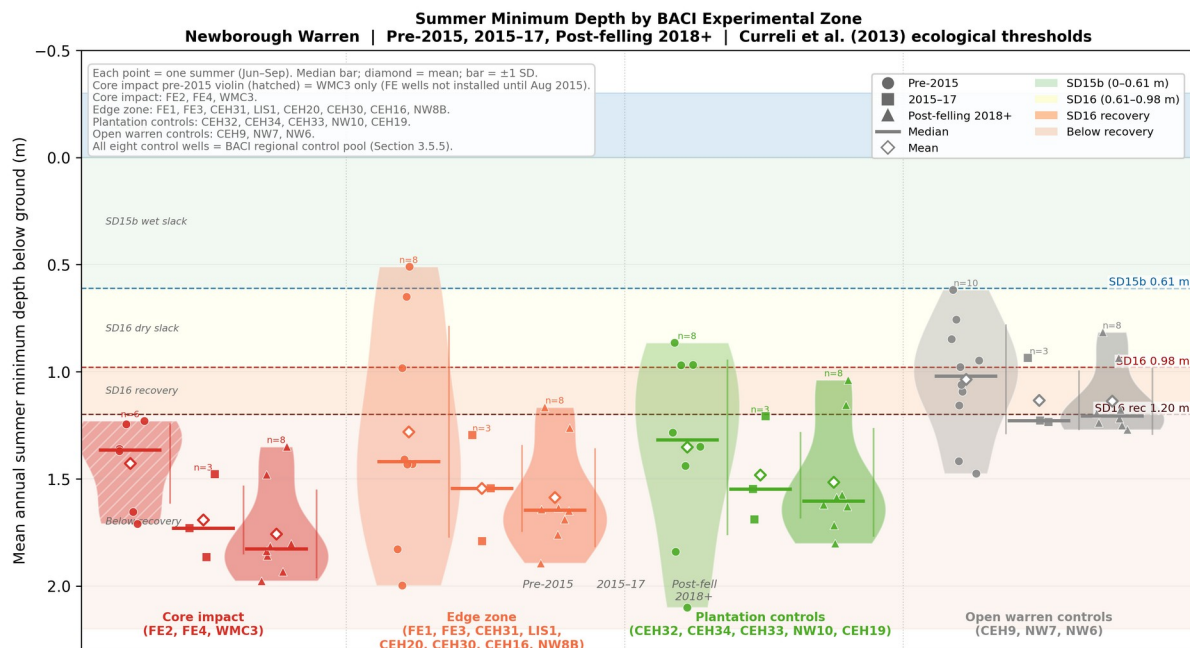


Figure 5. Summer minimum depth below ground surface by BACI experimental zone, Newborough Warren 2005–2026, across three phases (pre-2015 baseline, 2015–17 era, post-felling 2018+). Core impact: FE2, FE4, WMC3 (within the clearfell footprint); edge zone: FE1, FE3, CEH31, LIS1, CEH20, CEH30, CEH16, NW8b (forest margin); forest interior: CEH2, CEH13, CEH32, CEH33, CEH34 (uncleared); regional control: CEH19, NW10. Each point is one summer (June–September); ecological zone colouring follows Curreli et al. (2013). All zones sit below the SD16 recovery threshold (1.20 m) throughout the monitoring period. Hollingham (2026), Figure 25.

In net terms, the monitoring evidence suggests that clearance as implemented in December 2017 has been unhelpful for dune slack hydrology in the area that was felled. Whether this generalises to other plantation blocks, other scales of clearance, or other intervention designs is an open question; the 2017 felling was 8.4 ha of a particular block on the ridge-to-sea drainage pathway, and is not a clean test of wider or differently configured canopy management.

5.2 Dune scraping: a maintenance tool, not a cure, and not hydrologically neutral

Three scraping interventions were analysed: CEH36 (April 2015), CEH18 (October 2023), and CEH21 (October 2023). CEH36 returned a net climate-corrected benefit of +0.143 m during the pure scraping era, independently corroborated by synthetic control analysis (−0.116 m gap with donor wells). The benefit was partially eroded following the December 2017 clearfell (post-felling net benefit −0.047 m). CEH18 showed a measurable improvement following its 2023 scraping (era mean 0.950 m vs post-felling baseline 1.223 m), while CEH21 showed limited response.

A less obvious consequence of scraping is that a scraped hollow acts as a drainage sink for the area upslope of it: water that would otherwise have sustained the upslope water table now has somewhere lower to drain to. At CEH36 that upslope area includes the plantation block

subsequently felled in December 2017. This is why the clearfell analysis is careful to separate the scraping step (-0.137 m) from the felling step (-0.093 m); the two interventions were not hydrologically independent. Scraping is therefore best understood as a local intervention with non-local consequences — useful for making the scraped slack wetter, but redistributing water from its surroundings.

The Tier 1 CUSUM at the unmanipulated coastal control CEH4 shows progressive deterioration relative to the regional mean (-0.096 m) over the scraping period, consistent with coastal drainage dominating its water balance independently of any scraping signal. The overall picture is that scraping slows deterioration at treated sites but does not reverse the background climate-driven trajectory, and is most effective where summer water table floors can be raised rather than just winter peaks captured.

Tier 2 - Pure Scraping Signal (Paired CUSUM Analysis)

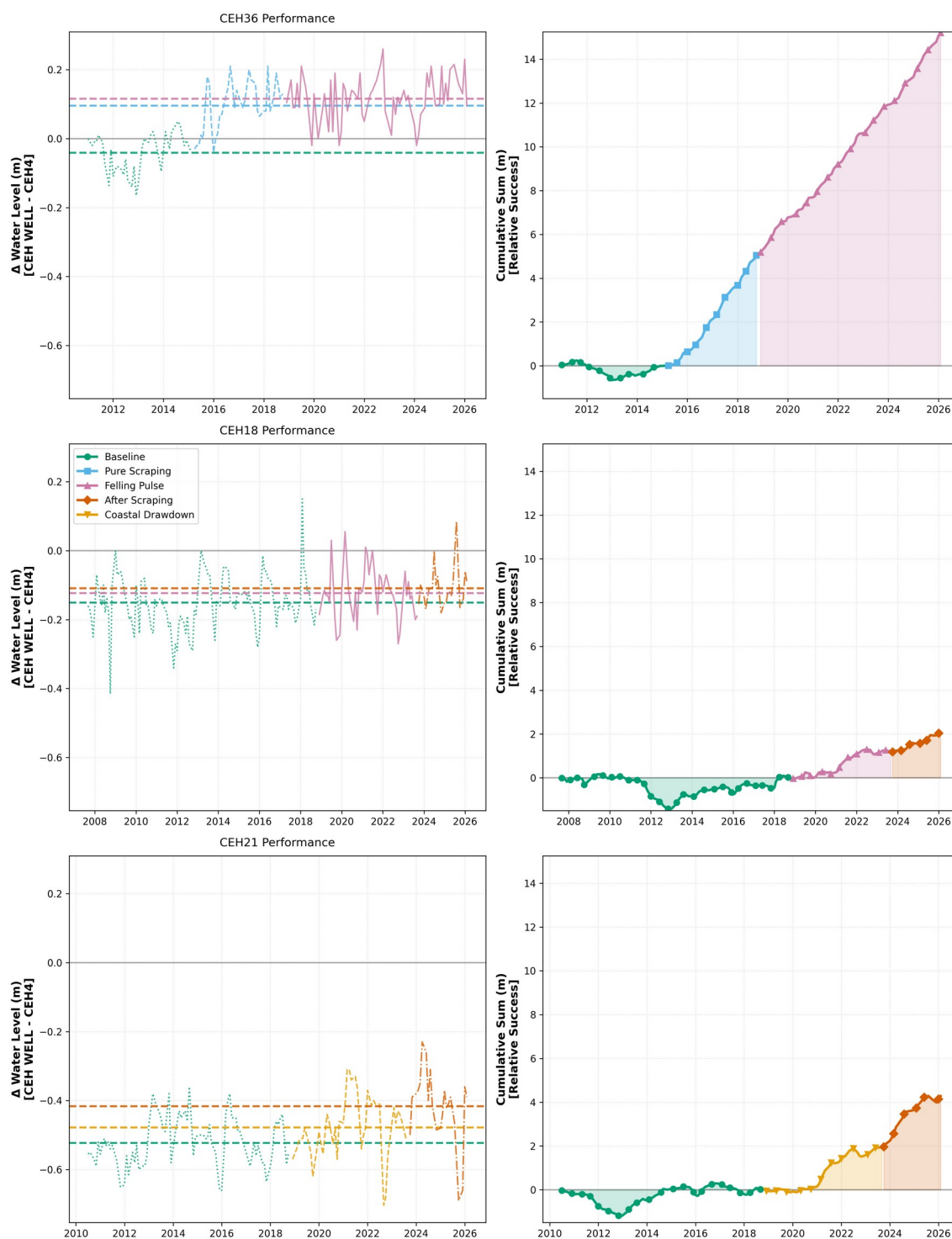


Figure 6. Tier 2 paired CUSUM analysis at scraping treatment wells CEH36 (top), CEH18 (centre) and CEH21 (bottom), differenced against matched controls (CEH36 and CEH18 vs CEH4; CEH21 vs CEH22). Left panels: monthly Δ water level relative to control with era means. Right panels: cumulative BACI CUSUM — rising indicates the treatment well becoming shallower relative to control. CEH36's rising CUSUM post-2018 reflects CEH4's deterioration as much as CEH36 improvement; CEH18 sits consistently negative relative to CEH4 with only a modest 2023 recovery; CEH21's rising post-2022 CUSUM should be interpreted cautiously as CEH22 is itself deteriorating through coastal erosion. Hollingham (2026), Figure 17.

6. The system-wide summer decline

Annual summer minimum water table depths are deepening at statistically significant rates in every structural cluster. Winter maxima show no significant trend anywhere.

Projected Climate Trajectory vs Ecological/Flooding Thresholds: Newborough Warren

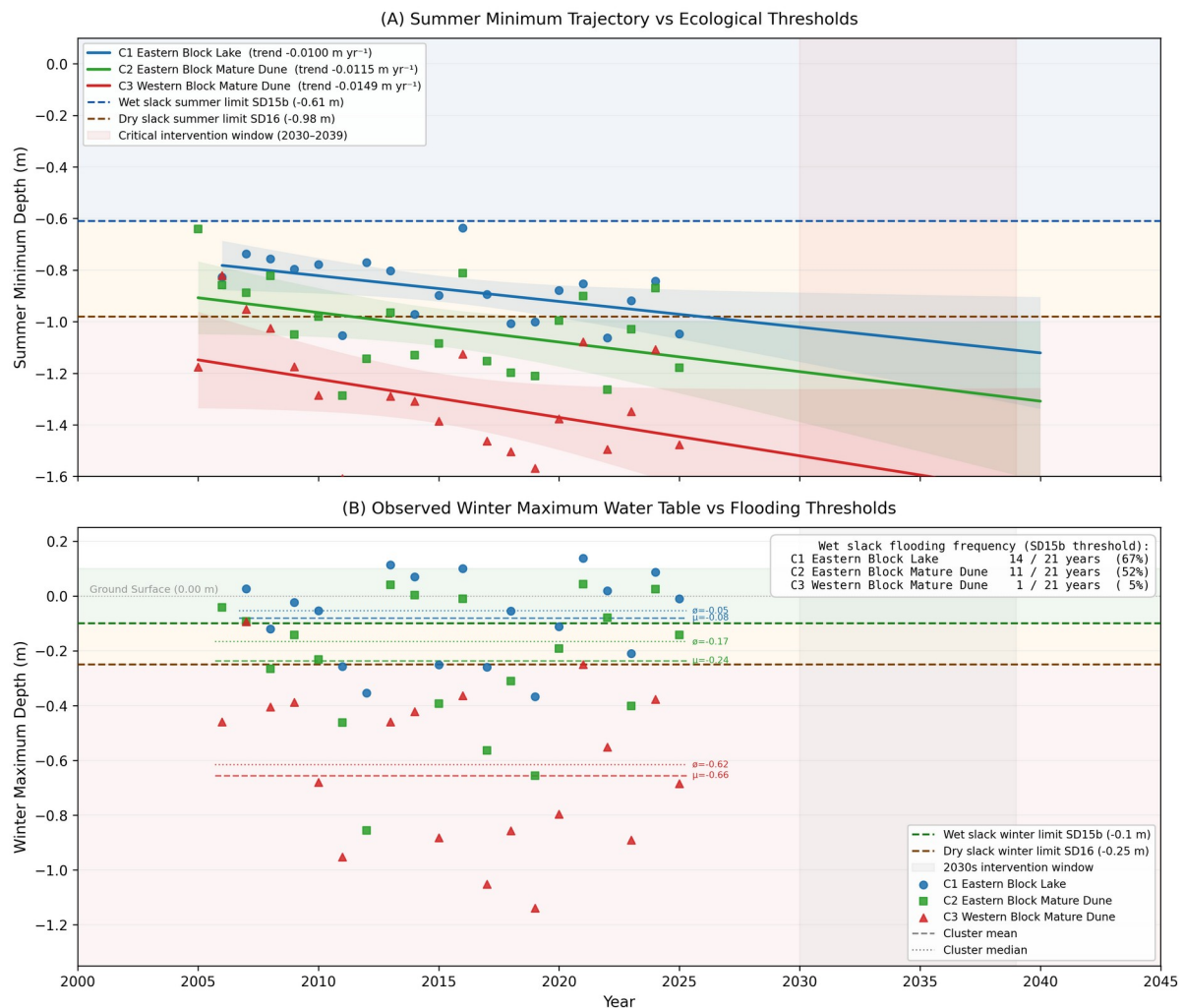


Figure 7. Climate trajectory and threshold exceedance analysis, Newborough Warren 2005–2025 with projection to 2040 (Hollingham 2026, Figure 30). Upper panel: summer minimum trajectory by cluster against Curreli et al. (2013) ecological thresholds — C1 -0.010 m/yr, C2 -0.012 m/yr, C3 -0.015 m/yr. Lower panel: observed winter maximum water tables against flooding thresholds (no significant winter trend in any cluster). The near-identical summer decline rates across clusters with very different canopy cover, substrate and drainage pathways identify a single site-wide forcing; the intervention window through the 2030s is shaded.

The spatial coherence of this signal — consistent across clusters that are hydrogeologically distinct, with different canopy cover and substrate, and crucially on separate drainage systems — is the key constraint. C1 sits on a drainage system that runs eastward to its own section of coastline and is hydrologically independent of the forest; for progressive forest evapotranspiration to be the site-wide driver, it would need to reach clusters the forest cannot reach. The more parsimonious reading is that the forest exerts a real local effect on its own

footprint and ridge-to-sea drainage pathway, and that a separate, site-wide forcing is superimposed on top of it.

The most direct candidate for the site-wide component is rising summer evaporative demand. The RAF Valley climate record shows a statistically significant warming trend in summer maximum temperatures of $+0.014^{\circ}\text{C yr}^{-1}$ over the full 95-year record ($p < 0.001$), with the post-2013 period running 0.94°C above the pre-2013 baseline of 17.89°C ; 2022 ($+1.77^{\circ}\text{C}$ anomaly, 747 mm annual rainfall) is the most severe dual-stress year on record. Winter rainfall has increased marginally ($+4.75 \text{ mm/yr}$, $p = 0.049$) but the highly transmissive sandy aquifer drains this surplus rapidly, leaving summer minima increasingly exposed (Figure 8).

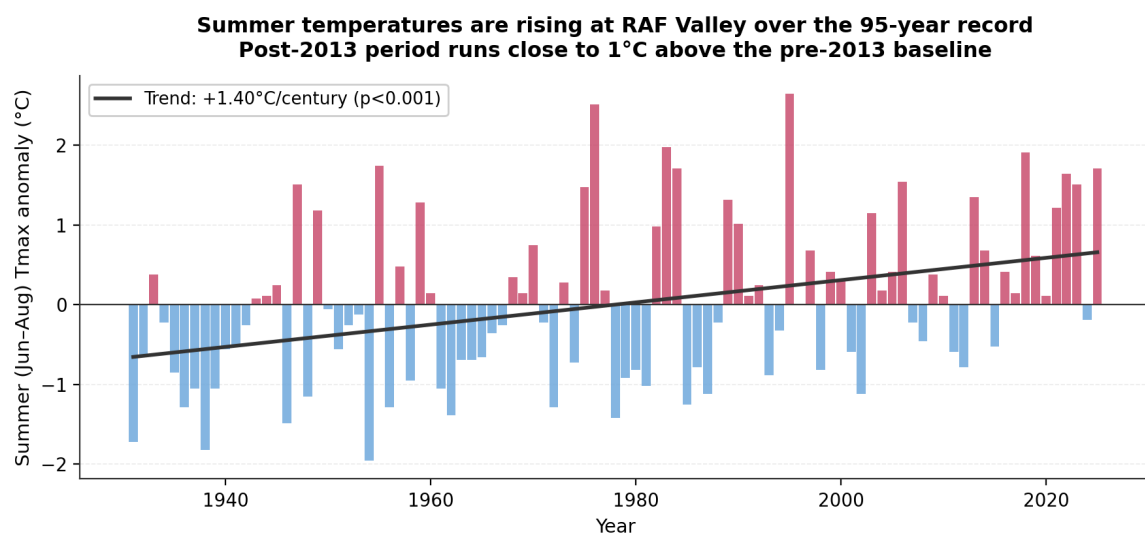


Figure 8. Summer temperature anomaly at RAF Valley (Met Office), 1930–2025. Red bars indicate summers warmer than the 95-year mean; blue bars indicate cooler summers. The trend line shows a statistically significant long-term warming of approximately 1.4°C per century ($p < 0.001$), with the post-2013 period running close to 1°C above the pre-2013 baseline.

A second candidate, not directly supported by the current monitoring evidence but not ruled out by it, is a long-run decline in the lateral inflow identified in Section 4. Separating this from the direct warming signal would require measurement on the ridge itself and on Llyn Rhos-Ddu. Both candidate drivers share a common upstream cause — a warmer summer climate — and the evidence for the direct evaporative-demand pathway is stronger than the evidence for the indirect lateral-inflow pathway.

7. P_flood and summer transfer functions: operational thresholds

The cluster-level SSM coefficients were algebraically inverted to produce critical rainfall threshold equations (P_{flood}) — the minimum monthly winter rainfall required to raise the water table to the slack floor from any given antecedent depth. These are the first operational management tools of their kind at Newborough Warren. The Western Block requires approximately 15–25% more monthly winter rainfall than the Eastern Block to close an equivalent deficit, consistent with its lower recharge sensitivity.

Peak flood transfer functions confirm that antecedent summer minimum depth is the dominant control on winter flooding severity in the Western Block ($R^2 = 0.684$, $p < 0.001$); winter rainfall amount alone is not a significant predictor ($p = 0.818$). This reframes the whole management problem: the dipwell readings most diagnostic of restoration trajectory are summer minima, not winter peaks.

The complementary direction — forecasting next summer's minimum depth from this winter's maximum and summer rainfall — is given by the following transfer functions (Hollingham 2026, Table 9):

Structural domain	Equation	R^2
Eastern Block (C1, C2)	$h_{\min} = 0.00093 \cdot P_{\text{summer}} + 0.333 \cdot h_{\max_winter} - 1.183$	0.41
Western Block (C3)	$h_{\min} = 0.00096 \cdot P_{\text{summer}} + 0.577 \cdot h_{\max_winter} - 1.214$	0.77
Forest (C4)	$h_{\min} = 0.00079 \cdot P_{\text{summer}} + 0.717 \cdot h_{\max_winter} - 1.180$	0.92

Table 2. Summer drought transfer functions for each structural domain. P_{summer} is total April–September rainfall (mm); $h_{\max_winter}$ and $h_{\min}$ are in metres below the surface (negative values). R^2 reflects the proportion of year-to-year variation in summer minimum depth that the equation explains.

Mapped against the eco-hydrological thresholds of Curreli et al. (2013), the current water table regime is already outside the range supporting wet dune slack communities across most of the monitored network. The Western Block (C3) summer minimum has fallen below the wet slack viability threshold in every monitored year and below the dry slack threshold in 81% of years (17 of 21); winter maxima exceeded the wet slack flooding threshold in only 1 of 21 monitored years (2007). The Eastern Block retains meaningful flooding capacity — C1 achieves wet slack winter flooding in 67% of years — but C2 is marginal at 52%, and at the current rate of summer minimum decline C1 will approach the dry slack summer threshold by the early 2030s.

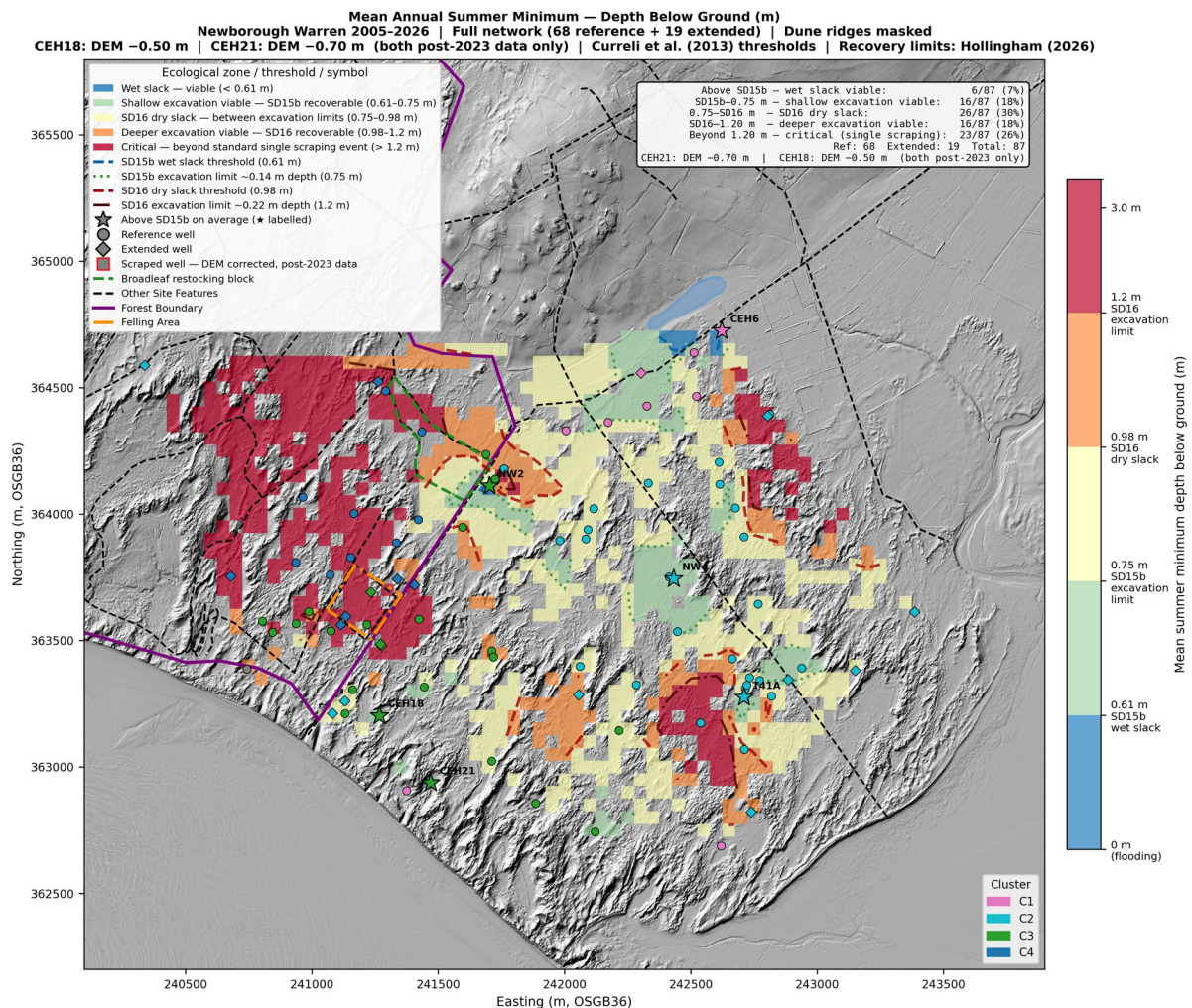


Figure 9. Spatial distribution of mean annual summer minimum water table depth below ground across the full dipwell network (68 reference + 19 extended wells), 2005–2026 (Hollingham 2026, Figure 26). Depth is metres below the LiDAR ground surface; the continuous IDW-interpolated surface is masked across dune ridges. Ecological zone boundaries follow Curreli et al. (2013): SD15b wet slack viability (0.61 m); SD15b excavation limit (0.75 m); SD16 dry slack threshold (0.98 m); SD16 excavation limit (1.20 m). The relatively shallow summer minima in the forest-adjacent zone (notably around NW2 and CEH2) reflect the structural position of these wells at the ridge recharge boundary rather than any beneficial effect of reduced canopy cover; clearance in this zone would be expected to deepen rather than raise summer minima. LiDAR data: © NRW and OS, Open Government Licence v3.0.

8. Clearfell: what the evidence does and does not support

The single most consequential management question concerns the hydrological consequences of plantation clearance. Three strands of evidence from the monitoring record bear on it:

- Summer minimum decline rates are statistically significant in every structural cluster and near-identical across canopy-free and forested clusters (C1: –0.010; C2: –0.012; C3: –0.015; C4: –0.017 m/yr). If progressive forest evapotranspiration were the dominant site-wide driver, open-dune clusters on drainage systems independent of the forest should not be declining at comparable rates. They are.

- The atmospheric draw coefficient β_2 increased at all 13 paired BACI wells after felling, confirming that canopy removal deepens summer water table drawdown in the area that was felled.
- Summer minimum depth is the dominant control on subsequent winter flooding in the Western Block — winter rainfall amount alone does not compensate for a deep summer deficit (Section 7).

Taken together, the evidence indicates that clearance as implemented in the December 2017 experiment has been unhelpful for dune slack hydrology in the area that was felled: the impact-zone water table stepped down at felling, continued to accumulate a climate-corrected deficit through February 2026, and the β_2 increase that accompanied canopy removal is the mechanism by which summer minima are deepened. The canopy provides a sacrificial evaporative shielding function that a cleared surface does not.

This is not the same as saying clearance at any scale or configuration would be counterproductive. Two caveats sit alongside the 2017 result. First, 8.4 ha is a small area on a particular block of the ridge-to-sea drainage pathway; what wider or differently designed canopy management might do is not resolved by a single intervention. Second, the ridge slopes themselves are forested, and if their interception is consuming water that would otherwise reach the dune slacks below, removing that interception could in principle increase ridge subsidy to the dune field. The 2017 felling was in effect a partial test of that argument, and the downstream slacks did not become wetter — but that is one test on one pathway, not a closed question.

Silvicultural management of canopy density — reducing stand density whilst retaining canopy structure — is identified in the main report as a more calibratable hydrological intervention than clearfell: stand density rather than species composition is the key management variable, and a more open canopy at reduced stand density would in principle recover much of the recharge deficit whilst retaining the shielding, buffering and drainage-attenuation functions the current closed stand provides. This remains a modelled expectation rather than a field-tested outcome.

Observational evidence from the 1993 clearfell and 1996 broadleaf restocking on the northern plantation margin — monitored through a single dipwell (NW10) over an 18-year shared record — suggests, subject to the positional confounders inherent in a one-well comparison, that broadleaf conversion may produce deeper summer minima than adjacent Corsican pine at comparable elevation. The implied mechanism is two-phase rather than single: during the deciduous dormant phase, reduced canopy interception allows greater throughfall to the aquifer, producing more winter recharge and more winter drainage; during the leaf-out phase, the larger summer leaf area increases evapotranspiration demand. The summer minimum is then deepened by both halves, and expresses as a larger seasonal amplitude rather than a higher β_2 (which is essentially identical between broadleaf and pine wells). An emerging deepening trend as the canopy closes (-28.8 mm/yr, $p = 0.042$, 2019–2025) indicates that the full hydrological cost of a mature closed broadleaf canopy may not yet be expressed in the current record. This is indicative rather than conclusive evidence and warrants replication across multiple broadleaf wells before any species-level conclusion is drawn.

9. Specific yield and aquifer uniformity

Water table fluctuation analysis returned cluster median S_y estimates of 0.22–0.26 across the four principal clusters after the Forest cluster was corrected for 24% canopy interception (Freeman, 2008). The corrected Forest value (0.227) is statistically indistinguishable from the open dune clusters, confirming that the plantation's distinctive hydrological signature is a surface boundary effect rather than a property of the sand.

These WTF-derived values are treated as empirical upper bounds; the 0.08 (C1) and 0.12 (C2–C4) values used in the primary volumetric water balance are retained as conservative lower bounds, reflecting the known tendency of monthly WTF to overestimate effective S_y through conflation of gravity drainage with capillary fringe release. Direct field measurement by slug test at two to three representative wells per cluster is identified as a priority for future work.

10. Limitations

Several limitations warrant particular attention:

- **Monthly resolution dampens event-scale volatility.** The derived parameters are optimised for seasonal trend analysis; daily event-scale peaks are smoothed out, so the assessment of seasonal amplitude is conservative.
- **The NW10 broadleaf comparison is a single uncontrolled well.** Observational evidence from NW10 is indicative rather than confirmatory — replication across multiple broadleaf wells and blocks would be needed for a statistically robust species effect.
- **The water balance residual cannot be mechanistically decomposed from water-level data alone.** Two independent diagnostic tests of the residual's time structure returned null results. Ridge-derived lateral recharge remains the most parsimonious single candidate, and the spatial distribution of positive boundary intercepts is aligned with the ridge boundary, but the residual is equally compatible with nonlinear soil-moisture dynamics and β -coefficient miscalibration the linear SSM does not capture. Direct measurement at the source candidates is needed to separate these alternatives.
- **The spatial model is not yet properly calibrated.** Hydraulic conductivity is constrained by a single tracer test ($K = 6.0$ m/day) and aquifer thickness in the western interior is inferred rather than measured. Forest management scenario difference maps are therefore not reliably distinguishable from interpolation artefacts; the spatial model yields informative results only for climate scenarios where all wells are affected simultaneously.

11. What comes next

The analytical framework is a robust foundation for the next generation of site management tools, but realising its full potential requires targeted field investment. Three measurements would materially strengthen the evidence base:

- **Slug tests at two to three representative wells per cluster** — would replace the single tracer-test K value with cluster-specific hydraulic conductivities, and provide direct Sy measurements to replace the WTF-based and literature-based bounds used in the volumetric water balance.
- **A ground-penetrating radar survey of aquifer thickness** — would replace inferred thicknesses with a mapped surface, enabling explicit modelling of the ridge recharge pathway that currently operates as an unmeasured boundary term.
- **Direct measurement at the candidate source locations of the water balance residual** — a water balance of Llyn Rhos-Ddu, a rain gauge on the ridge crest with a short piezometer transect on the ridge slopes, and spot gauging of any surface flow pathways crossing into the dune field — would allow the residual to be attributed to its actual physical sources rather than inferred from intercept audits.

Together these would allow the scenario difference maps to move from indicative to defensible, and would put the ridge-subsidy hypothesis on direct rather than circumstantial evidence. The P_flood equations and SSM framework have meanwhile been implemented in the Newborough Warren Hydrogeological Modelling Suite scenario viewer (Hollingham 2026b), a companion browser-based tool for exploring water table positions against flooding thresholds under projected climate forcing.

12. Bottom line for management

Summer minima, not winter rainfall, are the key ecological variable. Dune slack communities depend on the water table staying shallow through summer; summer minima also determine whether subsequent winter rainfall can restore slack flooding. Management should prioritise interventions that raise summer floors.

The site-wide deterioration is climate-driven. The forest exerts a real local effect on its own footprint and drainage pathway, but cannot explain the parallel decline in clusters on independent drainage systems. Site management can slow individual-slack deterioration but cannot reverse the underlying forcing.

Framing the challenge as adaptation, not restoration. Western dune slacks appear to be structurally beyond the reach of realistic winter rainfall; eastern slacks remain viable but on a closing intervention window. Tracking summer minima and responding year by year is more honest than targeting a historical baseline that the climate may no longer permit.

References

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Where to find out more

The full technical report, analysis code, data and supporting materials are available at:

https://github.com/newbroman/Newborough_Hydrology

Groundwater monitoring data were collected independently by the author. Climate data are from the RAF Valley Met Office station under the Open Government Licence. Topographic data are from Natural Resources Wales LiDAR under the Open Government Licence v3.0.